



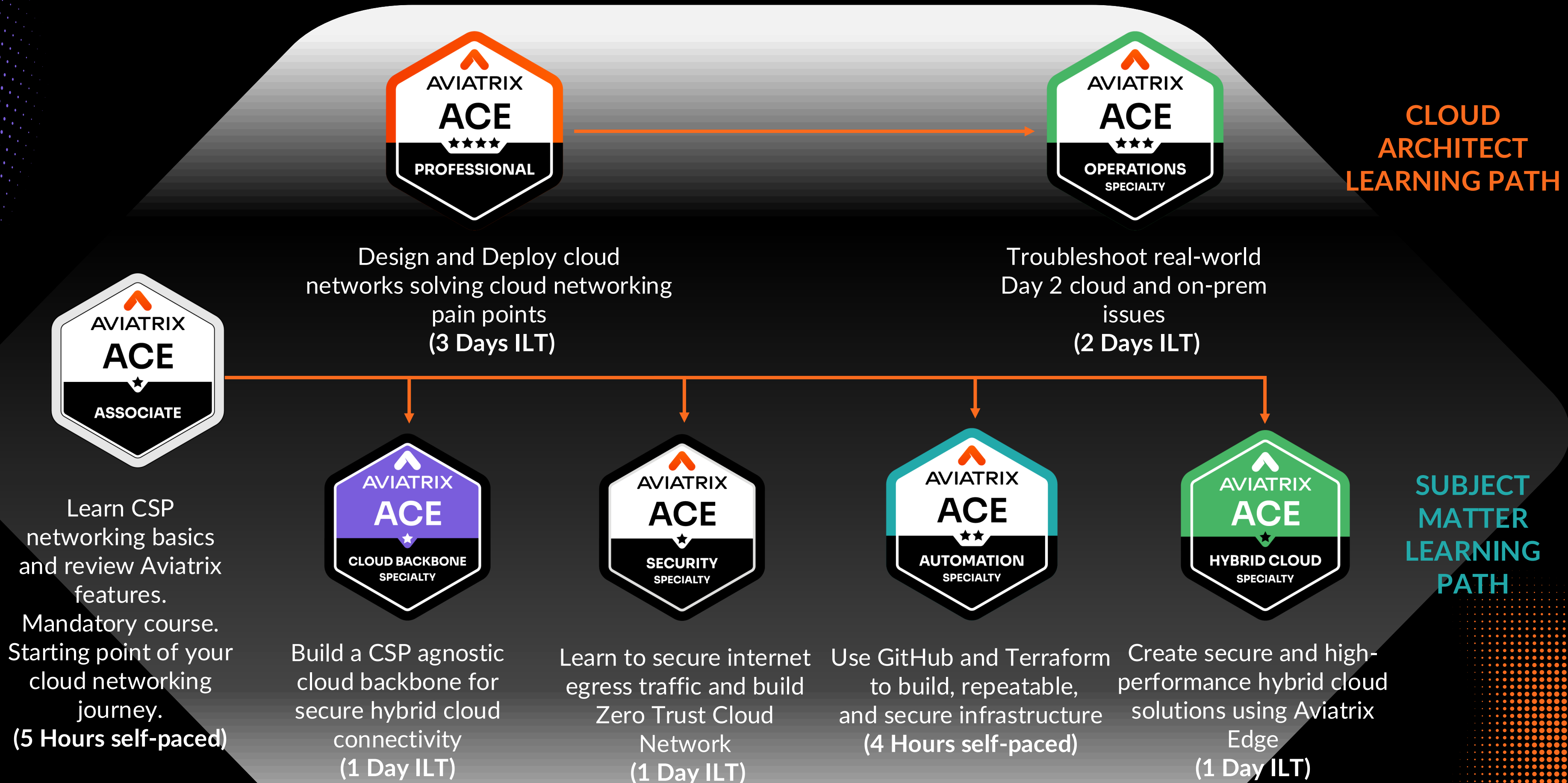
# ACE Professional

Day-1

ACE Team



# ACE Learning Paths



ILT = Instructor Led Training

# ACE Professional – Prerequisite and How to get certified

## Prerequisite:

- ACE Associate

## Certification Requirements:

To achieve certification, you must fulfil the following criteria:

- 1.Lab Completion:** Successfully complete at least **80%** of the labs, which equates to approximately 9 out of 12 labs. An automated script will assess the completion status of each lab.
  - 2.Course Attendance:** Full attendance is mandatory. Missing any class session will be considered a "no-show" for the entire course.
  - 3.Final Survey Completion:** You are required to complete a brief final survey, which should take no more than 3 minutes.
- By adhering to these guidelines, you will be well on your way to obtaining your certification.
  - The certification will be issued approximately seven business days after the final day of training.
  - Recordings are not to be shared. However, PDFs of the slides used during the bootcamp will be distributed. are not to be

# The ACE Associate is a mandatory step

- You are welcome to attend the training, but the ACE Associate certificate is a must to get the ACE-Pro certificate.
- If after successfully finishing this training, should you proceed to attain ACE-Associate certification, kindly reach out to [ace@aviatrix.com](mailto:ace@aviatrix.com) to request your ACE-Pro certificate.

*It is a manual process and can take a minimum 5 business days*

# LAB POD Policy

- ❑ If Day 1 registrations are complete but no labs are completed, the pod will be decommissioned. Attendees arriving on Day 2 may participate in the training without the Lab Pod and will not receive the ACE-Pro Certificate.
- ❑ Day 1 attendance is mandatory. If you miss Day 1, no pods will be available on Day 2. Attendees arriving on Day 2 may participate in the training without the Lab Pod and will not receive the ACE-Pro Certificate.



# Warm-Up (very important)

<https://docs.aviatrixlab.com/ace-pro/docs/home.html>



Welcome to ACE Professional Lab

POD Portal

Lab 1 - VPCs/VNets CREATION

Lab 2 - TRANSIT NETWORKING

Lab 3 - NETWORK SEGMENTATION

Lab 4 - HPE WITH ACTIVE MESH

Lab 5 - AVIATRIX CLOUD FIREWALL (with Secure Egress)

Lab 6 - FIRENET

Lab 7 - SITE2CLOUD

Lab 8 - SECURE HIGH-PERFORMANCE DATACENTER EDGE

Lab 9 - THREAT PREVENTION

Lab 10 - COSTIQ

Lab 11 - DISTRIBUTED CLOUD FIREWALL

Lab 12 - IAC & NETWORK INSIGHTS API

LOGOS-ICONS

PDFs

POD Portal

Link

You can view your personal POD portal by clicking on the picture below.

Access Code: 5RCY



ACE Professional Portal

Giuseppe Amendolara

Register

New Class

List

Status

Auto (system default)

Light

Dark

Logout

You have been assigned: Pod 1

Aviatrix CoPilot

Username: student

Password: 322z5dFuzY#R3Dp

Open Copilot

AWS

Username: student

Password: 322z5dFuzY#R3Dp

AWS Console

Gatus

Username: student

Password: 322z5dFuzY#R3Dp

aws-us-east-1-spoke1-test1

aws-us-east-1-spoke1-test2

aws-us-east-2-spoke1-test1

aws-us-east-2-spoke1-test2

azure-west-us-spoke1-test1

azure-west-us-spoke2-test1

gcp-us-central1-spoke1-test1

aws-strongswan-csr

onprem-pod1-edge

# Agenda – Day 1

| Topics                                         | Approx. Duration | Start time      |
|------------------------------------------------|------------------|-----------------|
| Welcome / Logistics / Agenda / Intro / Warm-Up | 0:50             | 9:00 AM         |
| CoPilot UI tour                                | 1:00             | 9:50 AM         |
| <b>Break</b>                                   | <b>0:15</b>      | <b>10:50 AM</b> |
| <b>LAB 1:</b> Create VPCs/VNet using CoPilot   | 0:30             | 11:05 AM        |
| CNSF - Characteristics                         | 0:20             | 11:35 AM        |
| CNSF - Deployment                              | 0:50             | 11:55 AM        |
| <b>Lunch Break</b>                             | <b>1:00</b>      | <b>12:45 PM</b> |
| <b>LAB 2:</b> CNSF                             | 0:55             | 1:45 PM         |
| Network Segmentation                           | 0:30             | 2:40 PM         |
| <b>LAB 3:</b> Network Segmentation             | 0:30             | 3:10 PM         |
| <b>Break</b>                                   | <b>0:15</b>      | <b>3:40 PM</b>  |
| High-Performance Encryption - HPE              | 0:25             | 3:55 PM         |
| ActiveMesh, Failover and Control Plane         | 0:40             | 4:20 AM         |
| <b>End</b>                                     |                  | <b>5:00 PM</b>  |

**Note:** Schedule might change if modules are running longer than expected.

# Agenda – Day 2

| Topics                           | Approx. Duration | Start time      |
|----------------------------------|------------------|-----------------|
| Day 1 Recap (Quiz)               | 0:20             | 9:00 AM         |
| LAB 4: HPE with ActiveMesh       | 0:50             | 9:20 AM         |
| Security                         | 0:50             | 10:10 AM        |
| <b>Break</b>                     | <b>0:15</b>      | <b>11:00 AM</b> |
| LAB 5: Aviatrix Cloud Firewall   | 0:50             | 11:15 AM        |
| Firewall Network - FireNet       | 0:45             | 12:05 PM        |
| <b>Lunch Break</b>               | <b>1:00</b>      | <b>12:50 PM</b> |
| LAB 6: Firewall Network with PAN | 0:55             | 1:50 PM         |
| Site2Cloud                       | 1:00             | 2:45 PM         |
| <b>Break</b>                     | <b>0:15</b>      | <b>3:45 PM</b>  |
| LAB 7: Site2Cloud                | 0:30             | 4:00 PM         |
| LAB 8: Aviatrix Edge             | 0:30             | 4:30 AM         |
| <b>END</b>                       |                  | <b>5:00 PM</b>  |

**Note:** Schedule might change if modules are running longer than expected.

# Agenda – Day 3

| Topics                                      | Approx. Duration | Start time      |
|---------------------------------------------|------------------|-----------------|
| Day 2 Recap (Quiz)                          | 0:20             | 9:00 AM         |
| ThreatQ & CostIQ                            | 0:20             | 9:20 AM         |
| <b>LAB 9</b> : Threat Prevention            | 0:30             | 9:40 AM         |
| <b>LAB 10</b> : CostIQ                      | 0:20             | 10:10 AM        |
| Distributed Cloud Firewall                  | 0:50             | 10:40 AM        |
| <b>Break</b>                                | <b>0:15</b>      | <b>11:30 AM</b> |
| <b>LAB 11:</b> Distributed Cloud Firewall   | 1:15             | 11:45 AM        |
| IaC and Network Insights API                | 0:20             | 1:00 PM         |
| <b>Lunch Break</b>                          | <b>1:00</b>      | <b>1:20 PM</b>  |
| <b>LAB 12:</b> IaC and Network Insights API | 0:25             | 2:20 PM         |
| Design Exercise Recap                       | 0:20             | 2:45 PM         |
| <b>Break</b>                                | <b>0:15</b>      | <b>3:00 PM</b>  |
| User VPN                                    | 0:30             | 3:15 PM         |
| Roadmap and new Features                    | 0:30             | 3:45 PM         |
| Survey + Closing                            | 0:10             | 4:15 PM         |
| <b>End</b>                                  |                  | <b>4:25 PM</b>  |

**Note:** Schedule might change if modules are running longer than expected.



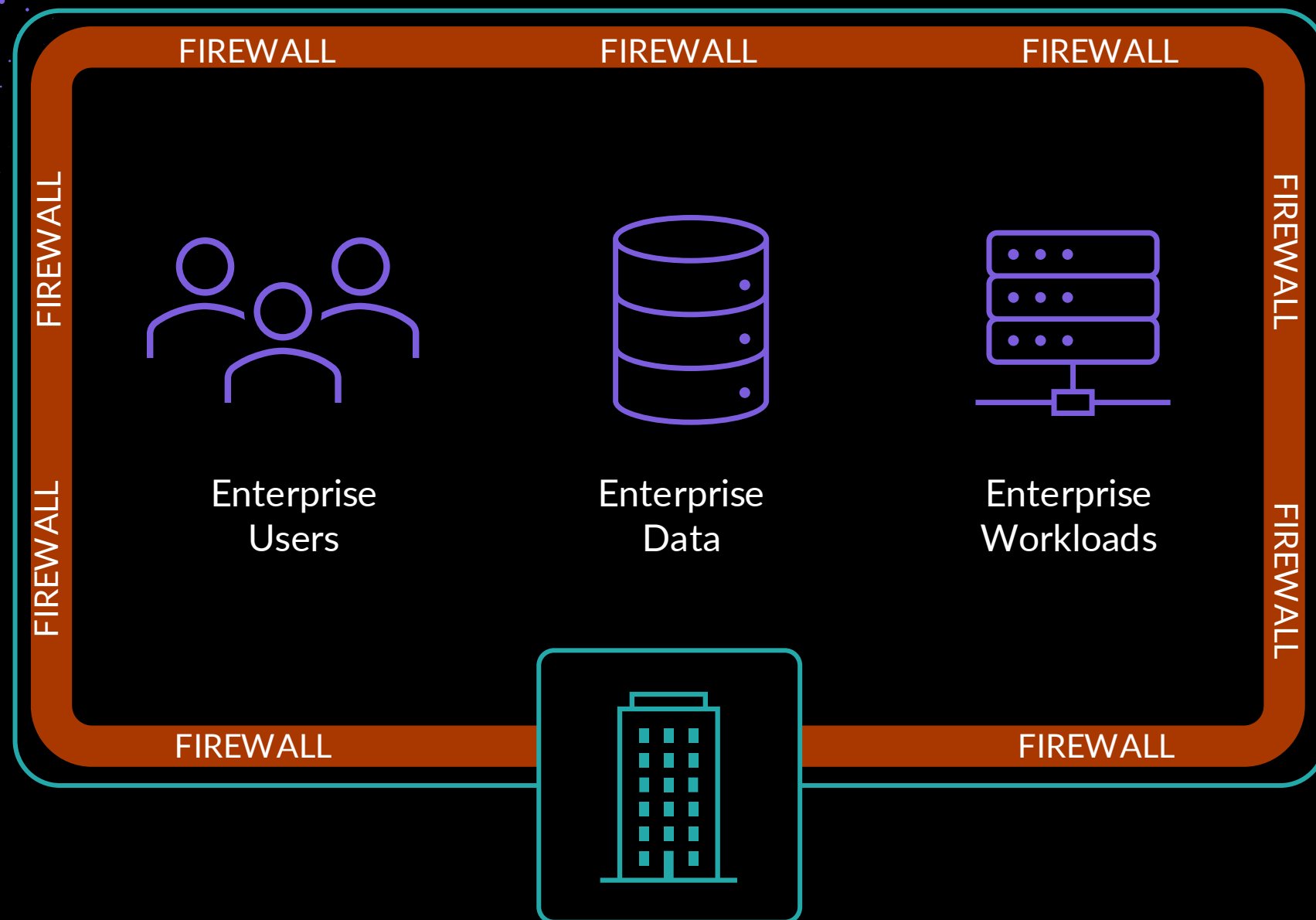
Who we are? What we do?

# The Cloud Security Challenge

*SECURING THE NEW ATOMIZED CLOUD PERIMETER*

## The Enterprise Network

A physical building with a cable to the Internet



## The Cloud Network

A highly dynamic and distributed environment



### The New Perimeter

Every Virtual Private Cloud (VPC), Kubernetes cluster, serverless function, PaaS service, and API endpoint

# Anatomy of Today's Threats

*RECENT SUCCESSFUL ATTACKS ARE FOLLOWING THE SAME PATTERNS*

Identity Compromise

Lateral Movement

Total Chaos

1

Initial access through stolen credentials

2

Undetected lateral movement for 9 days

3

Ransomware payload and data exfiltration

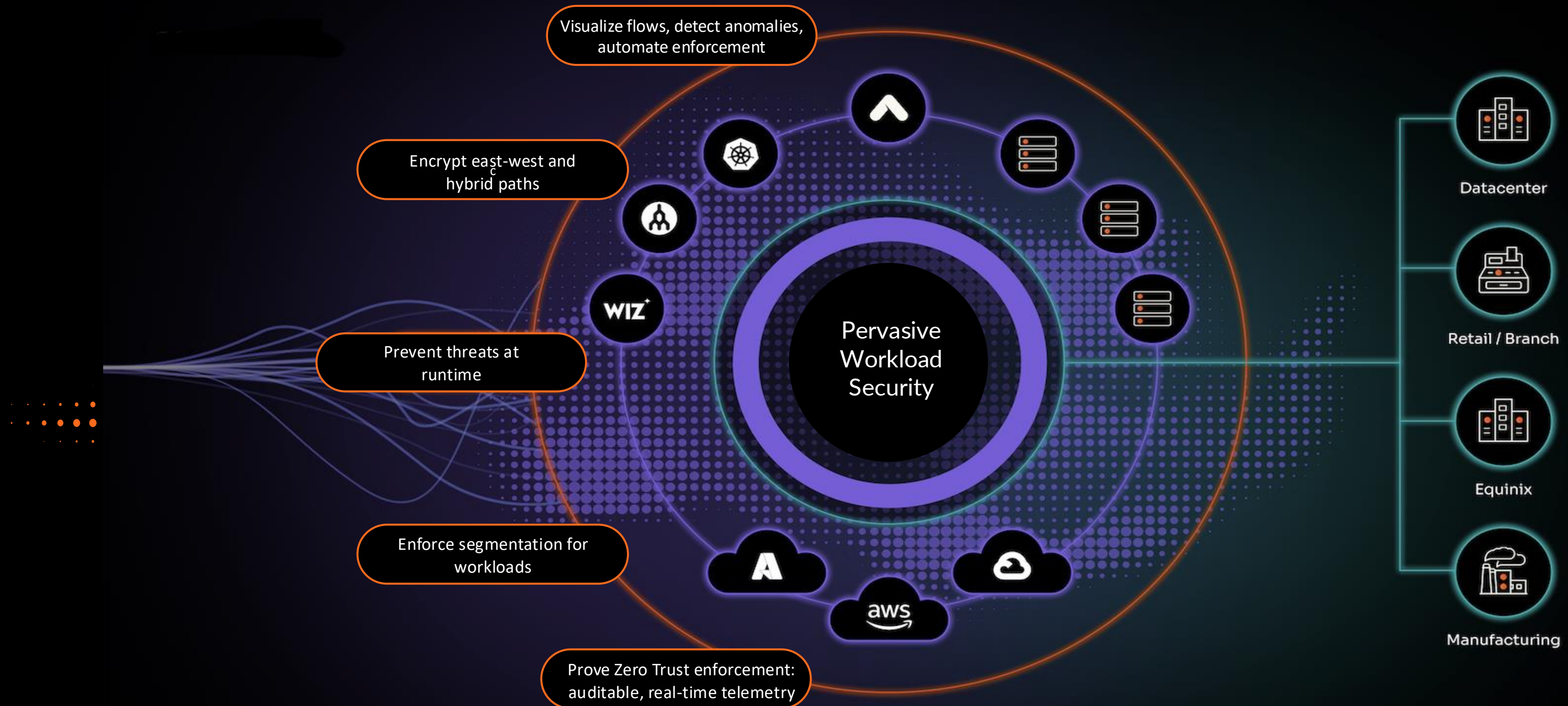


## UHG Operational Disruption and Impact

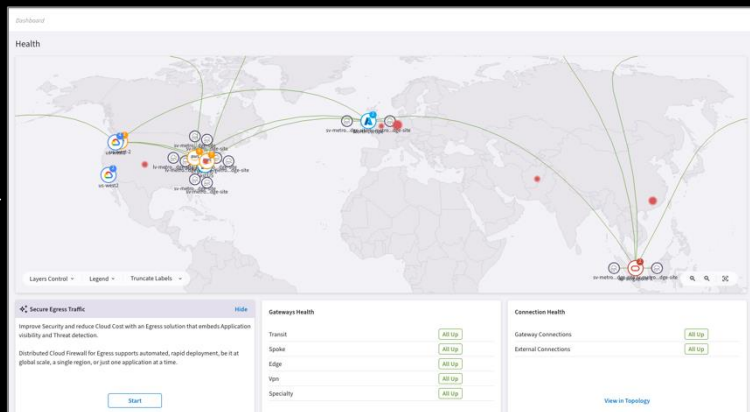
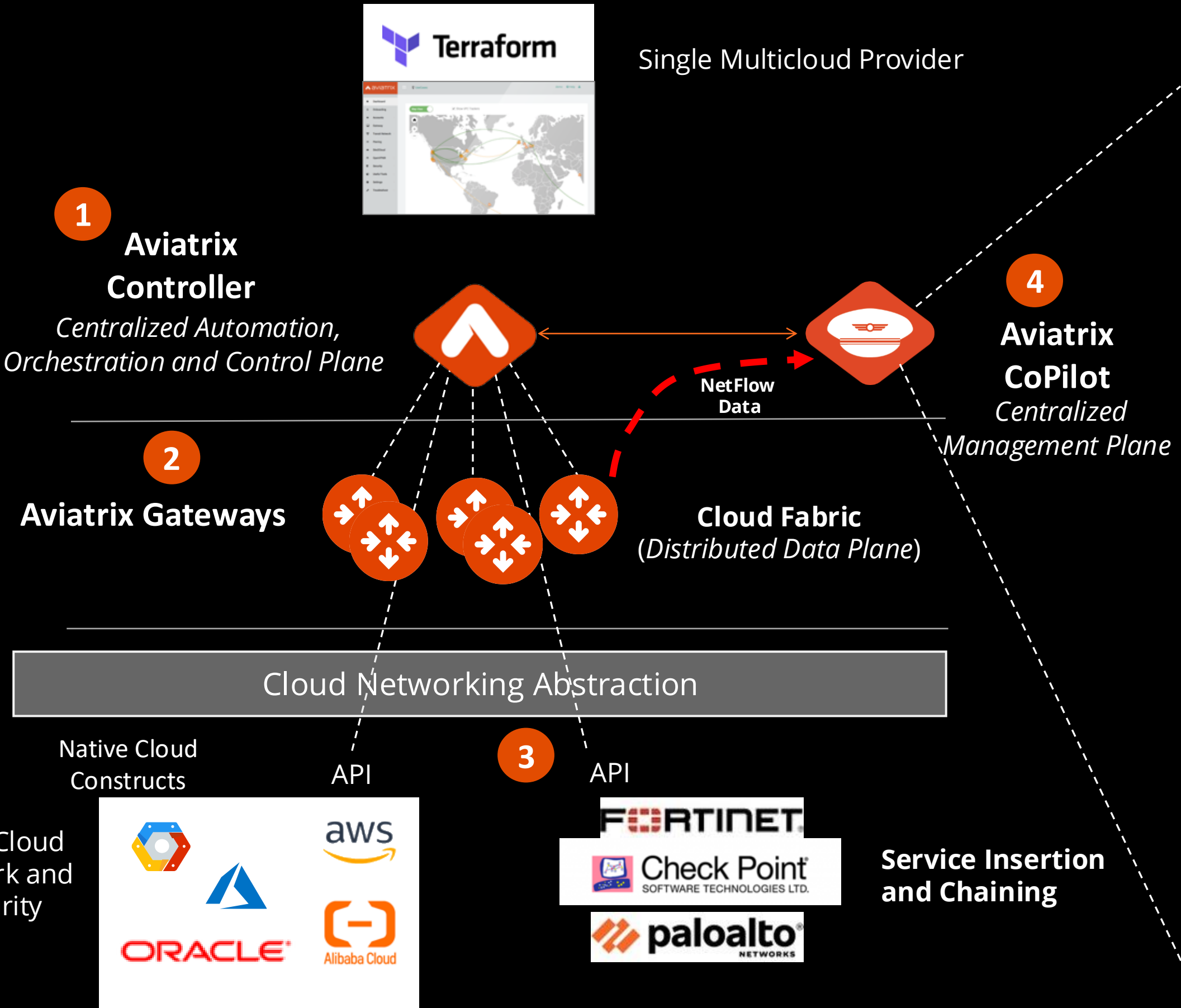
- \$22M ransom paid in Bitcoin — yet stolen data remains in the hands of threat actors, signaling a failed containment
- 190 million Americans exposed, including full identity profiles (names, DOBs, SSNs, driver's license numbers) — the largest healthcare breach in U.S. history
- Nationwide halt in claims processing disrupted patient care and provider payments for weeks — forcing Congressional intervention and federal investigation
- \$3.96B in damages reported — including \$3.09B in remediation and \$867M in operational losses, with long-tail legal and reputational fallout still unfolding

# The Solution – “Security From The Inside Out”

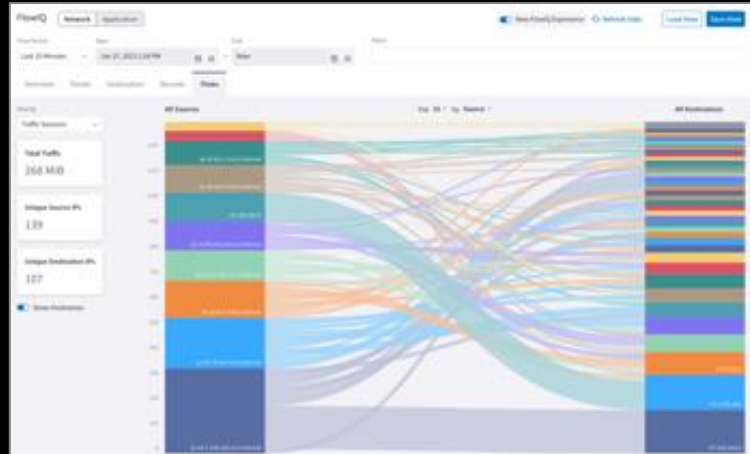
*CLOUD NATIVE SECURITY: EMBEDDED, NOT BOLTED ON*



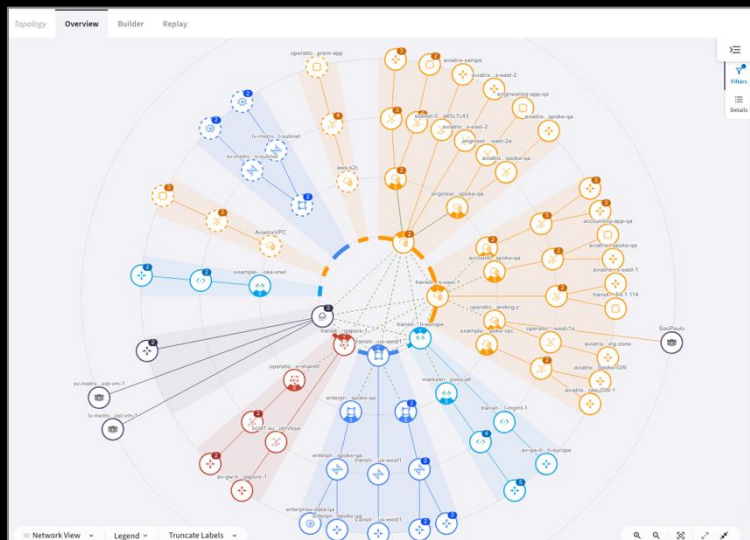
# CNSF (Aviatrix Enterprise)



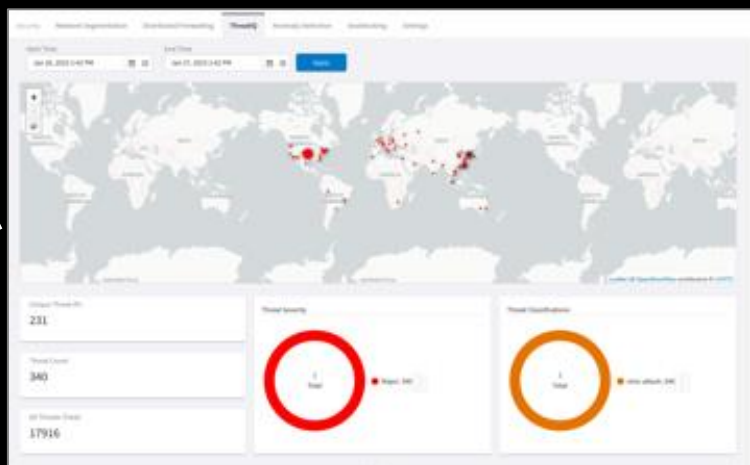
Multicloud Operational Visibility



FlowIQ Multicloud Traffic Flow Analysis



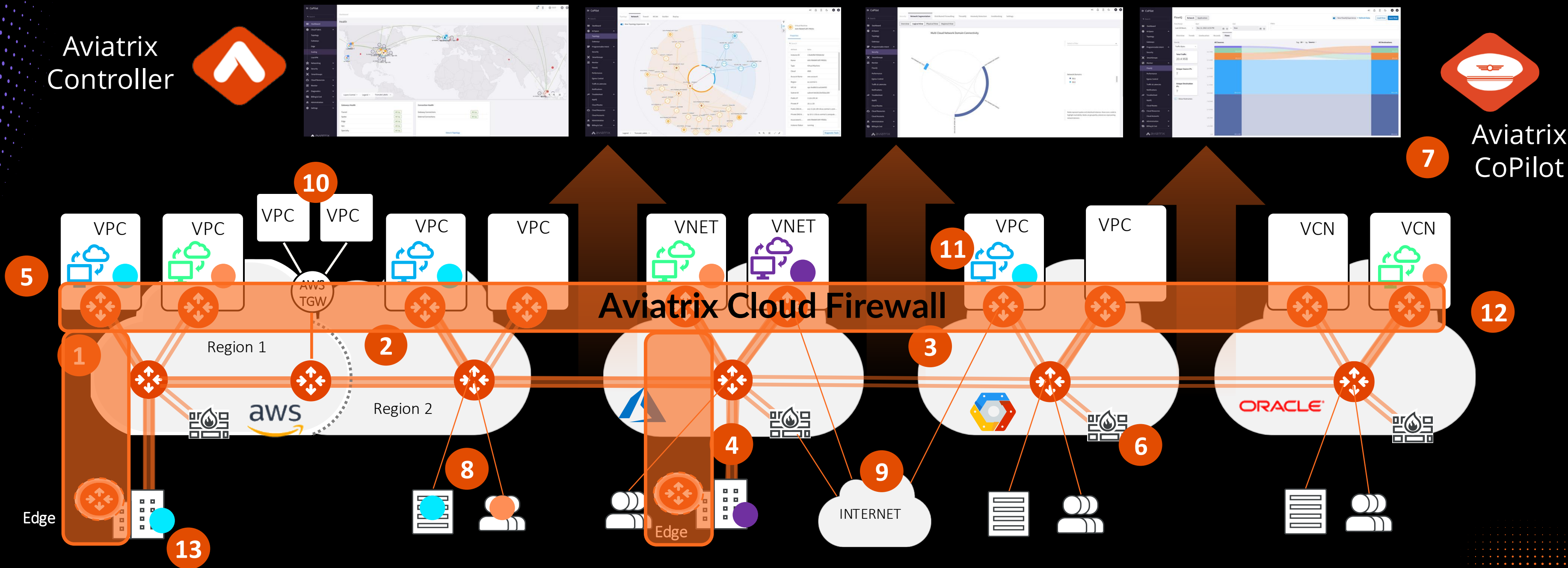
Multicloud Dynamic Topology Mapping



Threat Prevention

There are two products: **Aviatrix Enterprise** and **Aviatrix PaaS**

# CNSF – Logical View



1. Single Cloud  
Multi-Account  
High-Availability (Active-Active)  
Secure Hybrid-Cloud Connectivity

2. Multi-Region

3. Repeatable Design

4. High-Performance Encryption

5. Network Segmentation  
(aka Macro-Segmentation)

6. Service Insertion & Chaining

7. Enterprise Operational Visibility

8. Secure Cloud Access  
(User VPN)

9. Aviatrix Cloud Firewall  
(Secure Cloud Egress)

10. Cloud-Native

11. Micro-Segmentation

12. Distributed Cloud Firewall

13. Secure High-Performance  
Datacenter Edge

|                                                                                                     |                                                                                                          |                                                                                                        |                                                                                                                              |                                                                                                                   |                                                                                                                 |                                                                                                                 |                                                                                                           |                                                                                                                         |                                                                                                           |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|  <b>Abbott</b>     |  <b>abbvie</b>          |  <b>Adobe</b>         |  <b>Aflac</b>                             |  <b>ABInBev</b>                |  <b>APPTIO</b>               |                              |  <b>AVIS</b><br>Budget |  <b>BAKER HUGHES</b><br>a GE company |  <b>BankUnited</b>     |
|  <b>BHP</b>        |  <b>Biogen</b>          |  <b>BNY MELLON</b>   |  <b>CHOICE</b><br>HOTELS                  |  <b>CENTENE</b><br>Corporation |  <b>CHANGE</b><br>HEALTHCARE |  <b>chewy.com</b>            |  <b>CHS</b>            |  <b>Constellation</b><br>Brands      |  <b>covetrus</b>       |
|  <b>Cummins</b>    |  <b>DELL</b>            |  <b>dōTERRA</b>      |  <b>DXC</b><br>TECHNOLOGY                 |  <b>e-on</b>                   |  <b>FACTSET</b>              |  <b>FEMA</b>                 |  <b>Ford</b>           |  <b>Genworth</b>                     |  <b>GN</b>             |
|  <b>GUIDEWIRE</b> |  <b>Heineken</b>       |  <b>Home Office</b> |  <b>IHG</b>                              |  <b>Informatica</b>           |  <b>inmarsat</b>            |  <b>Johnson</b><br>Controls |  <b>Jefferies</b>     |  <b>Lufthansa</b><br>Systems        |  <b>lululemon</b>     |
|  <b>LG</b>       |  <b>Mercedes-Benz</b> |  <b>NASA</b>        |  <b>navitaire</b><br>an amadeus company |  <b>ni</b>                   |  <b>NIKE</b>               |  <b>PACCAR</b>             |  <b>pwc</b>          |  <b>Raytheon</b><br>Technologies   |  <b>RheinEnergie</b> |
|  <b>Roche</b>    |  <b>salesforce</b>    |  <b>SIEMENS</b>    |  <b>splunk</b>                          |  <b>Takeda</b>               |  <b>THALES</b>             |  <b>UNITED</b><br>AIRLINES |  <b>VERINT</b>       |  <b>WIX</b>                        |  <b>YARA</b>         |

# Next: Walkthrough of CoPilot UI

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